

PATTERSON, LA, USA

WMO: 722329

Lat: 29.717N

Lon: 91.333W

Elev: 3

StdP: 101.29

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12936

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.6	(c) 2.0	(d) -14.0	(e) 1.1	(f) 13.0	(g) -11.1	(h) 1.4	(i) 15.5	(j) 10.2	(k) 16.6	(l) 9.1	(m) 16.2	(n) 2.8	(o) 20	(p) 0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	33.9	24.5	32.9	24.9	32.4	24.9	27.4	31.5	26.9	30.9	26.5	30.5	3.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	21.8	29.5	26.0	21.3	29.2	25.4	20.6	28.6	86.6	31.6	84.4	31.0	82.3	30.4	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.5	7.7	7.0	DB	-3.1	35.3	3.0	2.0	-5.3	36.7	-7.1	37.9	-8.8	39.0	-11.0	40.5	
			WB	-5.2	28.0	2.5	1.2	-7.0	28.9	-8.4	29.6	-9.8	30.3	-11.7	31.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.6	12.0	14.2	17.2	20.3	24.1	26.9	27.9	27.8	26.2	21.3	16.1	13.1
	DBStd	6.82	5.45	5.17	4.41	3.43	2.59	1.41	1.34	1.31	2.23	3.83	4.48	5.10
	HDD10.0	95	42	18	5	0	0	0	0	0	0	0	5	26
	HDD18.3	722	204	133	74	21	2	0	0	0	0	18	94	176
	CDD10.0	3974	105	134	227	309	438	507	554	553	487	349	189	122
	CDD18.3	1558	9	16	38	79	181	257	296	294	237	109	28	13
	CDH23.3	13384	13	38	134	467	1472	2462	2994	2863	1985	783	133	39
(13)	CDH26.7	4834	0	1	4	53	448	989	1233	1148	751	197	10	1
(14) Wind	WSAvg	3.0	3.6	3.7	3.6	3.5	3.1	2.5	2.2	2.0	2.4	2.7	3.0	3.3
(15) Precipitation	PrecAvg	1520	131	88	95	91	109	172	178	173	133	120	106	119
	PrecMax	2284	442	233	283	251	439	556	252	338	352	317	363	377
	PrecMin	1182	9	14	2	2	6	40	99	75	40	10	19	47
	PrecStd	293	109	66	63	70	95	108	41	69	63	83	86	77
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	26.3	27.4	29.0	32.6	34.0	36.2	35.2	35.2	31.4	27.8	26.2
		MCWB	20.4	21.2	21.1	21.6	20.4	23.0	19.5	26.7	26.2	24.1	21.8	21.5
	2%	DB	23.0	24.3	26.2	27.9	31.3	32.8	33.8	33.6	32.8	30.1	26.4	24.1
		MCWB	19.3	20.3	20.5	21.7	20.7	23.8	23.4	26.1	25.7	23.3	20.9	20.4
	5%	DB	22.2	22.9	25.0	27.3	30.2	32.2	32.8	32.7	32.1	28.8	25.0	22.8
		MCWB	19.1	19.4	20.1	21.4	22.3	24.3	24.6	25.8	25.3	22.5	20.0	19.2
	10%	DB	20.9	22.0	23.7	26.3	29.0	31.3	32.2	32.1	31.0	27.6	23.4	21.3
		MCWB	18.2	18.9	19.4	20.4	22.4	24.4	24.7	25.8	24.6	21.2	19.2	18.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	22.7	22.9	24.3	25.8	27.6	28.0	28.0	27.6	26.2	23.9	23.1
		MCDB	23.7	25.0	25.9	27.2	29.8	31.3	32.7	33.0	32.0	29.2	26.3	25.1
	2%	WB	20.7	21.6	21.9	23.6	25.2	26.9	27.3	27.5	26.8	25.0	22.8	21.7
		MCDB	22.6	23.4	24.3	26.6	28.9	30.5	31.5	31.6	30.8	28.4	25.1	23.2
	5%	WB	19.5	20.6	21.2	22.9	24.7	26.3	26.8	26.9	26.2	24.1	21.6	20.6
		MCDB	21.4	22.3	23.7	25.9	28.3	30.0	30.9	31.0	30.1	27.1	23.9	22.5
	10%	WB	18.2	19.4	20.4	22.1	24.1	25.7	26.3	26.4	25.6	23.1	20.3	19.0
		MCDB	19.9	21.2	23.0	24.9	27.6	29.6	30.3	30.4	29.4	25.8	22.9	20.8
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	9.5	9.8	9.9	9.4	8.6	8.3	8.4	8.8	10.6	10.9	10.0
		MCDBR	9.4	9.0	9.2	9.5	9.8	9.6	9.4	9.3	9.7	10.5	10.7	10.3
		MCWBR	6.8	5.9	5.3	5.1	4.5	4.2	4.5	4.1	4.2	5.3	6.7	7.5
	5% WB	MCDBR	8.3	8.0	8.0	8.0	8.0	8.1	8.4	8.6	8.5	8.6	9.2	9.1
		MCWBR	6.8	6.1	5.5	5.0	3.8	3.7	4.0	4.2	3.9	4.8	6.6	7.3
(40) Clear-Sky Solar Irradiance	taub		0.343	0.358	0.378	0.423	0.449	0.467	0.496	0.491	0.441	0.385	0.360	0.352
	taud		2.515	2.481	2.412	2.290	2.251	2.249	2.199	2.212	2.319	2.460	2.471	2.498
	Ebn at Noon		892	908	909	874	848	828	803	804	835	867	864	861
	Edh at Noon		89	100	114	134	139	139	146	143	124	101	92	87
(44) All-Sky Solar Radiation	RadAvg		3.00	3.55	4.81	5.82	6.39	6.20	5.79	5.47	5.13	4.58	3.46	2.69
(45)	RadStd		0.28	0.47	0.43	0.42	0.34	0.48	0.40	0.47	0.44	0.49	0.33	0.25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	+86

Nomenclature: See separate page